

Ratio, proportion and percentages - Set 1 - Solutions

Specification M3.1-M3.11.

Answer key

Question	Answer	Topic
1	D	Ratio, proportion and percentages
2	E	Ratio, proportion and percentages
3	B	Ratio, proportion and percentages
4	C	Ratio, proportion and percentages
5	F	Ratio, proportion and percentages
6	D	Ratio, proportion and percentages
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9	C	Ratio, proportion and percentages
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Worked solutions

Question 1 RAT-01 · Ratio, proportion and percentages · spec M3.8

The price of a coat is reduced by 20% in a sale, and then the sale price is reduced by a further 25%. The final price is what percentage of the original price?

Answer: D

$$0.8 \times 0.75 = 0.6, \text{ so } 60\%.$$

Full method - how and why

Successive percentage changes **multiply**. Convert each to a multiplier and combine.

- 1 A 20% reduction multiplies by 0.8.
why: You keep 80% of the price: $1 - 0.20 = 0.8$.
- 2 A further 25% reduction multiplies by 0.75 - applied to the **sale** price, not the original.
why: The second discount acts on whatever the price already is; that is why the changes multiply.
- 3 Combine: $0.8 \times 0.75 = 0.6$.
why: Multiplying the multipliers gives the overall multiplier.
- 4 So the final price is 60% of the original.
why: Equivalently a 40% reduction overall - not 45%.

Where it goes wrong

- Adding the percentages to get 45% off, hence 55% - an offered option.
- Applying the second reduction to the original price.

Takeaway Percentage changes compose by multiplying multipliers, never by adding percentages.

Question 2 RAT-02 · Ratio, proportion and percentages · spec M3.8

After a 15% increase, a salary is £27 600. What was the original salary?

Answer: E

$$1.15 \times S = 27600 \Rightarrow S = 24000.$$

Full method - how and why

A reverse-percentage problem: the given figure is the **result**, so divide by the multiplier.

- 1 A 15% increase means multiplying by 1.15.
why: $100\% + 15\% = 115\% = 1.15$.
- 2 Write the relation: $1.15 \times (\text{original}) = 27\,600$.
why: The original is the unknown; the multiplier acts on it.
- 3 Divide: $\text{original} = \frac{27\,600}{1.15} = 24\,000$.
why: $27600 \div 115 = 240$, so dividing by 1.15 gives 24 000.
- 4 Check: $24\,000 \times 1.15 = 27\,600$ ✓.
why: Always multiply back - it catches the commonest error instantly.

Where it goes wrong

- Taking 15% off the new salary: $27\,600 \times 0.85 = 23\,460$ - an offered option.
- Dividing by 0.15 or by 1.5.

Takeaway Reverse percentage: **divide** by the multiplier, then check by multiplying back.

Question 3 RAT-03 · Ratio, proportion and percentages · spec M3.4

£600 is divided between A , B and C in the ratio $2 : 3 : 7$. How much more does C receive than A ?

Answer: B

Each part is £50; $C - A = 5 \text{ parts} = £250$.

Full method - how and why

Sharing in a ratio: find the value of one part, then answer the question actually asked.

- 1 Add the ratio parts: $2 + 3 + 7 = 12$.

why: The total is divided into this many equal parts.

- 2 One part: $£600 \div 12 = £50$.

why: Divide the total by the number of parts.

- 3 The difference between C and A is $7 - 2 = 5$ parts.

why: Work with the **difference** in parts rather than computing both shares.

- 4 So the difference is $5 \times £50 = £250$.

why: Check: A gets £100, C gets £350 ✓.

Where it goes wrong

- Giving C 's share (£350) instead of the difference - an offered option.
- Dividing by 3 (the number of people) instead of 12 (the number of parts).

Takeaway Find the value of one part first; then read carefully whether a share, a difference or a total is wanted.

Question 4 RAT-04 · Ratio, proportion and percentages · spec M3.9

y is inversely proportional to the square of x . When $x = 2$, $y = 9$. Find y when $x = 6$.

Answer: C

$$y = \frac{k}{x^2}, k = 36. y = \frac{36}{36} = 1.$$

Full method - how and why

Inverse proportion to a **square**. Find the constant, then substitute.

- 1 Write the relationship: $y = \frac{k}{x^2}$.

why: 'Inversely proportional to the square of x ' puts x^2 in the denominator.

- 2 Find k from the given pair: $9 = \frac{k}{4}$, so $k = 36$.

why: Substitute $x = 2$, $y = 9$; note $x^2 = 4$, not 2.

- 3 Substitute $x = 6$: $y = \frac{36}{36} = 1$.

why: $6^2 = 36$.

Where it goes wrong

- Using $y = \frac{k}{x}$ and getting 3 - an offered option.
- Reasoning ' x triples so y thirds', which ignores the squaring (it should be a ninth).

Takeaway Inverse square: tripling x divides y by 9. Always find k explicitly.

Question 5 RAT-05 · Ratio, proportion and percentages · spec M3.11

A bacteria population doubles every 3 hours. By what factor does it grow in 24 hours?

Answer: F

$$24/3 = 8 \text{ doublings: } 2^8 = 256.$$

Full method - how and why

Repeated doubling. Count the doublings, then raise 2 to that power.

- 1 Count the doubling periods: $24 \div 3 = 8$.

why: Each 3-hour block doubles the population.

- 2 Each doubling multiplies by 2, so eight of them multiply by 2^8 .

why: Repeated multiplication is exponentiation - not multiplication by 8.

- 3 $2^8 = 256$.

why: Worth knowing the powers of 2 up to $2^{10} = 1024$.

Where it goes wrong

- Multiplying by 8 or computing $2 \times 8 = 16$ - both offered.
- Using $2^6 = 64$ by dividing 24 by 4.

Takeaway Growth factor = (factor per period)^{number of periods}.

Question 6 RAT-06 · Ratio, proportion and percentages · spec M3.10

Two similar solids have surface areas in the ratio 4 : 9. Find the ratio of their volumes.

Answer: D

$$\text{Length ratio } 2 : 3, \text{ so volume ratio } 2^3 : 3^3 = 8 : 27.$$

Full method - how and why

Similar solids: areas scale as the square of lengths, volumes as the cube. Work back to the length ratio first.

- 1 Areas are in ratio 4 : 9, so the length ratio is $\sqrt{4} : \sqrt{9} = 2 : 3$.

why: Area scales with the square of length, so undo it with a square root.

- 2 Volumes scale with the cube of length: $2^3 : 3^3$.

why: Every linear dimension is scaled, so three dimensions give the cube.

- 3 So the volume ratio is 8 : 27.

why: Check: the ratios 2 : 3, 4 : 9, 8 : 27 are the classic length/area/volume triple.

Where it goes wrong

- Cubing the **area** ratio to get 64 : 729 - an offered option.
- Leaving the answer as the length ratio 2 : 3.

Takeaway Length k , area k^2 , volume k^3 : always return to the length ratio before converting.

Question 7 RAT-07 · Ratio, proportion and percentages · spec M3.5

A mixture contains sand and cement in the ratio 5 : 2. How much cement must be added to 35 kg of the mixture to make the ratio 5 : 3?

Answer: E

35 kg contains 25 kg sand and 10 kg cement. For 5 : 3 with 25 kg sand, cement must be 15 kg: add 5 kg.

Full method - how and why

A changing ratio where one quantity stays fixed. Anchor on the unchanged amount.

- 1 Split the 35 kg in the ratio 5 : 2: seven parts of 5 kg, so 25 kg sand and 10 kg cement.

why: $35 \div 7 = 5$ kg per part.

- 2 Only cement is added, so the sand stays at 25 kg.

why: Identify what does **not** change - that is the anchor for the new ratio.

- 3 In the new ratio 5 : 3, the 5 parts correspond to 25 kg, so one part is 5 kg and the cement must be $3 \times 5 = 15$ kg.

why: Scale the new ratio to the fixed quantity.

- 4 Cement to add: $15 - 10 = 5$ kg.

why: The question asks for the amount added, not the final amount.

Where it goes wrong

- Answering 15 kg (the new total cement) - an offered option.
- Rescaling the sand as well, as though both quantities changed.

Takeaway In 'add to change the ratio' problems, anchor on the quantity that stays the same.

Question 8 RAT-10 · Ratio, proportion and percentages · spec M3.11

£2000 is invested at 5% compound interest per year. Which expression gives the amount after n years?

Answer: C

Compound growth multiplies by 1.05 each year.

Full method - how and why

Compound interest is repeated multiplication, which gives an exponential, not a linear, expression.

- 1 Each year the amount is multiplied by $1 + 0.05 = 1.05$.

why: Interest is added to the running balance, so next year's interest is larger.

- 2 After n years the original has been multiplied n times: 2000×1.05^n .

why: Repeated multiplication gives a power.

Where it goes wrong

- Choosing $2000 + 100n$ or $2000(1 + 0.05n)$ - those are **simple** interest.
- Choosing 2000×0.05^n , which would shrink the investment to almost nothing.

Takeaway Compound: $P(1 + r)^n$. Simple: $P(1 + rn)$. The word 'compound' means the multiplier repeats.

Question 9 RAT-13 · Ratio, proportion and percentages · spec M3.8

The value of a car falls by 20% each year. After how many complete years is it first worth less than half its original value?

Answer: C

$$0.8^3 = 0.512 > 0.5, 0.8^4 = 0.4096 < 0.5. \text{ So 4 years.}$$

Full method - how and why

Repeated decay compared with a threshold. Test successive powers - no logarithms needed.

- 1 Each year multiplies the value by 0.8.
why: A 20% fall leaves 80%.
- 2 After n years the multiplier is 0.8^n ; we need the first n with $0.8^n < 0.5$.
why: 'Less than half' is a condition on the multiplier.
- 3 Compute: $0.8^2 = 0.64$, $0.8^3 = 0.512$, $0.8^4 = 0.4096$.
why: Multiply successively rather than using logs - three multiplications is quick.
- 4 $0.8^3 = 0.512 > 0.5$ but $0.8^4 < 0.5$, so the answer is 4 complete years.
why: Note how close year 3 is - checking it is essential.

Where it goes wrong

- Stopping at 3 because 0.512 'is about a half'.
- Reasoning '20% per year, so half gone after 2.5 years' - decay is multiplicative.

Takeaway Test successive powers against the threshold, and always check the year before your answer.

Question 10 RAT-14 · Ratio, proportion and percentages · spec M3.5

A recipe for 12 biscuits uses 150 g of flour and 100 g of butter. Ann has 1 kg of flour and 500 g of butter. What is the greatest number of biscuits she can make?

Answer: B

Flour allows $\frac{1000}{150} \times 12 = 80$; butter allows $\frac{500}{100} \times 12 = 60$. Butter is the limit: 60.

Full method - how and why

A limiting-factor problem. Work out how many each ingredient allows, then take the smaller.

- 1 Flour: 1 kg = 1000 g, and 150 g makes 12 biscuits, so flour allows $\frac{1000}{150} \times 12 = 80$ biscuits.
why: Convert to the same units first.
- 2 Butter: 500 g at 100 g per 12 biscuits allows $\frac{500}{100} \times 12 = 60$ biscuits.
why: Same method for the second ingredient.
- 3 Take the **smaller**: butter runs out first, so 60 biscuits.
why: You can only make as many as the scarcest ingredient permits.

Where it goes wrong

- Taking the larger number (80) - an offered option.
- Averaging the two, or adding them.

Takeaway With several ingredients, the answer is the minimum over all of them - the limiting factor.

Question 11 RAT-15 · Ratio, proportion and percentages · spec M3.8

Which of the following statements is/are true?

- I Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.
- II A 10% increase followed by a 10% increase is a 21% increase overall.
- III If A is 25% more than B , then B is 20% less than A .

Answer: G

I is false: $1.5 \times 0.5 = 0.75$. II: $1.1^2 = 1.21$, true. III: $A = 1.25B \Rightarrow B = 0.8A$, true.

Full method - how and why

Three percentage claims. Each is settled by converting to multipliers.

- 1 I: +50% then -50% gives $1.5 \times 0.5 = 0.75$, i.e. 75% of the original. **False.**

why: The decrease acts on the larger amount, so it removes more than the increase added.

- 2 II: +10% twice gives $1.1^2 = 1.21$, a 21% increase. **True.**

why: The extra 1% is the interest on the first year's interest.

- 3 III: $A = 1.25B$ means $B = \frac{A}{1.25} = 0.8A$, which is 20% less than A . **True.**

why: Percentage changes are not symmetric because the base changes.

- 4 II and III only.

why: Match to the options.

Where it goes wrong

- Accepting I because the percentages 'cancel'.
- Rejecting III on the grounds that 25% up should mean 25% down.

Takeaway Always convert to multipliers: percentage changes are asymmetric because the base changes.

Question 12 RAT-08 · Ratio, proportion and percentages · spec M3.8

A quantity increases by $x\%$ and then decreases by $x\%$. The overall change is a decrease of 4%. Find x .

Answer: B

$$\left(1 + \frac{x}{100}\right)\left(1 - \frac{x}{100}\right) = 1 - \frac{x^2}{10000} = 0.96 \Rightarrow x^2 = 400 \Rightarrow x = 20.$$

Full method - how and why

An increase then an equal-percentage decrease. The multipliers form a difference of two squares.

- 1 Write the multipliers: $\left(1 + \frac{x}{100}\right)$ then $\left(1 - \frac{x}{100}\right)$.

why: Both act on the running value, so they multiply.

- 2 Multiply: $\left(1 + \frac{x}{100}\right)\left(1 - \frac{x}{100}\right) = 1 - \frac{x^2}{10000}$.

why: Difference of two squares - which shows the result is always less than 1.

- 3 A 4% overall decrease means the multiplier is 0.96: $1 - \frac{x^2}{10000} = 0.96$.

why: Convert the stated overall change into a multiplier.

- 4 Solve: $\frac{x^2}{10000} = 0.04 \Rightarrow x^2 = 400 \Rightarrow x = 20$.

why: Take the positive root, since x is a percentage size.

Where it goes wrong

- Assuming $x = 4$ because the net change is 4% - an offered option.
- Arithmetic: $0.04 \times 10000 = 400$, not 40.

Takeaway Up then down by $x\%$ always loses: the net multiplier is $1 - \frac{x^2}{10000}$.